Prüfung

MAT121 – Analysis I

7. Februar 2022

Aufgabe 1: (10 Punkte)

Entscheiden Sie für jede Aussage, ob sie wahr oder falsch ist. Falsche Antworten geben einen Minuspunkt, unbeantwortete Fragen geben keinen Abzug.

	wahr	falsch
i) Seien $A, B \subset \mathbb{Q}$ zwei Mengen, mit $a \leq b$ für alle $a \in A, b \in B$. Dann existiert $c \in \mathbb{Q}$ mit $a \leq c$ und $c \leq b$ für alle $a \in A, b \in B$.	☐	☒
ii) Die Menge, die aus allen Folgen $(x_n)_{n\in\mathbb{N}}$ besteht, mit $x_n \in \{0,1\}$ für alle $n \in \mathbb{N}$, ist überabzählbar.	☒	☐
* iii) Sei $A \subset \mathbb{R}$ beschränkt. Dann hat A ein eindeutiges Infimum in $\mathbb{R}$.	☒	☐
iv) Sei z_n eine Folge in $\mathbb{C}$ so, dass Re z_n und $\lvert z_n\rvert = \sqrt{\bar{z}_n z_n}$ konvergieren. Dann konvergiert auch z_n.	☐	☒
v) Sei $n \in \mathbb{N}$. Die Gleichung $z^n = i$ hat n verschiedene Lösungen in $\mathbb{C}$.	☒	☐
vi) Sei $f : \mathbb{R} \to \mathbb{R}$ analytisch. Dann ist $f \in C^\infty(\mathbb{R})$.	☒	☐
vii) Sei $f \in C^\infty(\mathbb{R})$ so, dass die Reihe $\sum_{n\geq 0} f^{(n)}(0)x^n/n!$ konvergiert, für alle $\lvert x\rvert < 1$. Dann ist f analytisch an der Stelle $x = 0$.	☐	☒
viii) Sei $f : [-1;1] \to \mathbb{R}$ stetig, differenzierbar auf $(-1;1)$. Sei $x_0 \in (-1;1)$ mit $f'(x_0) = 0$. Dann ist x_0 ein lokales Extremum von f.	☐	☒
ix) Sei $f : [0;1] \to \mathbb{R}$ beschränkt und auf $(0;1)$ differenzierbar. Dann ist f integrierbar auf $[0;1]$.	☒	☐
x) Sei $f : [0;1] \to \mathbb{R}$ integrierbar, $g : [0;1] \to \mathbb{R}$ beschränkt und so, dass $g(x) \leq f(x)$ für alle $x \in [0;1]$. Dann ist g auch auf $[0;1]$ integrierbar.	☐	☒

* Since I did not exclude $A = \emptyset$, I accepted both answers.

2) a) i) We expand

$$\cos x = 1 + \mathcal{O}(x^2), \quad \text{as } x \to 0$$

Thus

$$(n+1)\cos\left(\frac{1}{n+1}\right) - n\cos(1/n) =$$

$$= (n+1)\cdot\left(1 + \mathcal{O}\left(\frac{1}{n^2}\right)\right) - n\left(1 + \mathcal{O}(1/n^2)\right)$$

$$= 1 + \mathcal{O}(1/n)$$

We conclude

$$\lim_{n\to\infty}\left[(n+1)\cos\left(\frac{1}{n+1}\right) - n\cos(1/n)\right] = 1$$

ii) $n^{1/n} \longrightarrow 1 \Rightarrow n^{3/n} = \left(n^{1/n}\right)^3 \longrightarrow 1$

$$\frac{n^2-2}{3n^2+1} = \frac{1 - 2/n^2}{3 + 1/n^2} \longrightarrow \frac{1}{3}$$

Thus $\displaystyle\lim_{n\to\infty} \frac{n^2-2}{3n^2+1}\cdot n^{3/n} = 1/3$

b) Observe that

$n\cdot(n+1)/2$ is even if $\begin{cases} n = 4k & \text{or} \\ n = 4k+3 \end{cases}$ for a $k \in \mathbb{N}$.

$n\cdot(n+1)/2$ is odd if $\begin{cases} n = 4k+1 & \text{or} \\ n = 4k+2 \end{cases}$

Hence
$$a_n = \begin{cases} 1 & \text{if } n = 4k,\ 4k+3 \\ -1 & \text{if } n = 4k+1,\ 4k+2 \end{cases}$$

$\Rightarrow$ a_n has two accumulation points ± 1.

c) i) By L'Hopital:
$$\lim_{x \to 1} \frac{2^x - 2}{\log x} = \lim_{x \to 1} \frac{2^x \cdot \log 2}{1/x} = 2\log 2$$

ii) $\cos\sqrt{x} = 1 - \frac{x}{2} + o(x^2)$, as $x \downarrow 0$

$\Rightarrow$ $1 - \cos\sqrt{x} = x/2 + o(x^2)$, as $x \downarrow 0$

$\sin x = x + o(x^3)$, as $x \downarrow 0$

$$\Rightarrow \lim_{x \downarrow 0} \frac{1 - \cos\sqrt{x}}{\sin x} = \frac{1}{2}$$

iii) $\cos x = 1 - x^2/2 + \frac{x^4}{4!} + o(x^6)$

$\Rightarrow$ $1 - \cos x = x^2/2 - x^4/4! + o(x^6)$

$$\Rightarrow \frac{1}{1 - \cos x} = \frac{1}{\frac{x^2}{2}\left(1 - \frac{x^2}{12} + o(x^4)\right)}$$
$$= \frac{2}{x^2}\left(1 + \frac{x^2}{12} + o(x^4)\right)$$
$$= \frac{2}{x^2} + \frac{1}{6} + o(x^2)$$

$$\Rightarrow \lim_{x \to 0} \frac{1}{1 - \cos x} - \frac{2}{x^2} = 1/6$$

3) i) $x^n, x^{n+2} \xrightarrow[n\to\infty]{} 0 \quad \forall x \in [0;1)$

For $x=1$, $x^n - x^{n+2} = 0 \quad \forall n \in \mathbb{N}$

Thus $f_n(x) \longrightarrow 0$ pointwise.

We have $f_n'(x) = n \cdot x^{n-1} - (n+2)x^{n+1}$

$\Rightarrow f_n'(x) = 0$ if $x = \sqrt{\frac{n}{n+2}}$

Thus

$$\sup_{x\in[0;1]} |f_n(x)| = f_n\left(\sqrt{\frac{n}{n+2}}\right)$$

$$= \underbrace{\frac{2}{n+2}}_{\to 0} \cdot \underbrace{\left(1 - \frac{2}{n+2}\right)^{n/2}}_{\to e^{-1}} \xrightarrow[n\to\infty]{} 0$$

This implies that $f_n(x) \to 0$ uniformly.

ii) $f_n(x) = \cos(x - n\cdot\pi) = (-1)^n \cdot \cos x$

The sequence does not converge, not even pointwise (it converges only for $x = \pi/2 + k\cdot\pi$, $k \in \mathbb{N}$).

iii) For $x \neq 0$, we have

$$|f_n(x)| \leq \frac{nx}{n^2 x^4} \leq \frac{1}{n} \cdot \frac{1}{x^3} \xrightarrow[n\to\infty]{} 0$$

For $x = 0$, $f_n(x) = 0 \quad \forall n \in \mathbb{N}$.

Thus $f_n(x) \longrightarrow 0$ pointwise.

On the other hand, we have

$$f_n(1/\sqrt{n}) = \sqrt{n} \cdot \frac{1}{1+n^2 \cdot \frac{1}{n^2}} = \frac{\sqrt{n}}{2} \xrightarrow[n\to\infty]{} +\infty$$

Hence

$$\sup_{x\in[0,1]} |f_n(x)| \to +\infty \quad , \text{ as } n \to \infty$$

f_n does not converge uniformly.

b) i) Let $f_n(x) = n \cdot e^{-nx} \sin^2 x$.

We have

$$|f_n(x)| = n \cdot e^{-nx} |\sin x|^2 \leq$$

$$\leq n x^2 e^{-nx} = \frac{1}{n} \cdot n^2 x^2 e^{-nx}$$

$$\leq 1/n \cdot \sup_{y\in\mathbb{R}} y^2 e^{-y} \leq C/n$$

for all $x \in [0,1]$. Hence

$f_n(x) \to 0$ uniformly on $[0,1]$

$$\Rightarrow \lim_{n\to\infty} \int_0^1 n e^{-nx} \sin^2 x \, dx = \int_0^1 0 \cdot dx = 0$$

ii) $$\lim_{n\to\infty} \int_0^1 (3n+2) \cdot x^n \, dx = \lim_{n\to\infty} (3n+2) \cdot \int_0^1 x^n \, dx$$

$$= \lim_{n\to\infty} (3n+2) \cdot \frac{x^{n+1}}{n+1} \Big|_0^1 = \lim_{n\to\infty} \frac{3n+2}{n+1} = 3$$

(despite the fact that $(3n+2)x^n \to 0 \quad \forall x \in [0,1]$)

4) a) $f(0) = -1 < 0$

$f(\pi/2) = 1 - e^{-\pi/2} > 0$

(because $e^{\pi/2} > 1$)

Since f is continuous (as the difference of two continuous functions), the intermediate value theorem (Zwischenwertsatz) implies $\exists\, x_0 \in (0;\pi/2)$ with $f(x_0) = 0$.

Since

$f'(x) = \cos x + e^{-x} > 0 \quad \forall x \in [0;\pi/2]$

f is strictly monotone. Hence there can be only one zero in $[0;\pi/2]$.

b) Let $1 < y < x$. Then $\exists\, \lambda \in (0;1)$ with

$y = \lambda \cdot 1 + (1-\lambda) \cdot x = \lambda(1-x) + x$

Thus:

$$g(y) = g(\lambda \cdot 1 + (1-\lambda) \cdot x) = \frac{f(\lambda \cdot 1 + (1-\lambda) \cdot x)}{(x-1)(1-\lambda)}$$

$$\underset{\substack{\uparrow \\ \text{Konvexity of } f.}}{\leq} \frac{\lambda \cdot f(1) + (1-\lambda) f(x)}{(x-1)(1-\lambda)} = \frac{f(x)}{x-1} = g(x)$$

$\blacksquare$

5) a) i) $\left| (-1)^k \cdot \frac{3k^2+1}{5k^4+2k^2+6} \right| \leq \frac{4}{5} \cdot \frac{1}{k^2}$

Since

$$\sum_{k=1}^{\infty} 1/k^2 < +\infty ,$$

the series under consideration converges absolutely.

ii) $\frac{1}{\log n} \longrightarrow 0$ monotonically.

By Leibnitz, it follows that

$$\sum_{n=2}^{\infty} (-1)^n \frac{1}{\log n} \quad \text{converges}$$

On the other hand

$$\left| (-1)^n \cdot \frac{1}{\log n} \right| = \frac{1}{\log n} \geq \frac{1}{\sqrt{n}}$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = +\infty$$

the series under consideration does not converge absolutely.

iii) $\cos(n\pi) = (-1)^n$

$$\Rightarrow \quad 2^{n\cos(n\pi)} = \begin{cases} 2^n, & \text{if } n \text{ even} \\ 2^{-n}, & \text{if } n \text{ odd} \end{cases}$$

Dividing the sum in even and odd terms, we find

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + 2^{n\cos(n\pi)}} = \sum_{k=1}^{\infty} \frac{1}{\sqrt{2k} + 2^{2k}} + \sum_{k=1}^{\infty} \frac{1}{\sqrt{2k+1} + 2^{-2k-1}}$$

The first sum converges, since

$$\frac{1}{\sqrt{2k} + 2^{2k}} \leq \frac{1}{2^{2k}} = \frac{1}{4^k}$$

and $\sum_{k=1}^{\infty} 1/4^k < +\infty$. But the second sum diverges, since

$$\frac{1}{\sqrt{2k+1} + 2^{-2k-1}} \geq \frac{1}{2\sqrt{2k+1}} \geq \frac{1}{4\sqrt{k}}$$

and $\sum_{k=1}^{\infty} 1/\sqrt{k} = +\infty$.

Hence, the sum under consideration diverges.

b) i) We have

$$\rho = \frac{1}{\lim_{k\to\infty} \left[(k^{1/k}-1)^k\right]^{1/k}} = \frac{1}{\lim_{k\to\infty} (k^{1/k}-1)} = +\infty$$

ii) We find

$$\rho = \lim_{n\to\infty} \frac{n!\,5^n}{n^n} \cdot \frac{(n+1)^{n+1}}{(n+1)!\,5^{n+1}} = \lim_{n\to\infty} \frac{1}{5}\left(\frac{n+1}{n}\right)^n$$

$$= \lim_{n\to\infty} \frac{1}{5}\left(1+\frac{1}{n}\right)^n = e/5$$

6) a) We proceed by contradiction.

Suppose f is not injective.

Then $\exists x_1, x_2 \in [a;b]$ with $x_1 < x_2$ and

$$f(x_1) = f(x_2) = c$$

Define $g : [x_1; x_2] \to \mathbb{R}$ by

$$g(x) = f(x) - c$$

Then: g is continuous, diff'ble on $(x_1; x_2)$

and $$g(x_1) = g(x_2) = 0$$

The lemma of Rolle implies that

$\exists \xi \in (x_1; x_2) \subset [a;b]$ with

$$g'(\xi) = f'(\xi) = 0$$

in contradiction with the assumption. $\blacksquare$

b) <u>Step 1</u>: $\exists y_1 \in (x_1; x_2)$ with $f'(y_1) < 0$.

<u>Proof</u>: if $f'(\xi) \geq 0 \quad \forall \xi \in (x_1; x_2)$, then

f would be increasing and

$$f(x_2) \geq f(x_1)$$

in contradiction with assumption.

<u>Step 2</u>: $\exists y_2 \in (x_2; x_3)$ with $f'(y_2) > 0$.

<u>Proof</u>: if $f'(\xi) \leq 0 \quad \forall\, \xi \in (x_2; x_3)$, then f would be decreasing and

$$f(x_3) \leq f(x_2)$$

in contradiction with the assumption.

<u>Step 3</u>: $\exists\, z \in (y_1; y_2)$ with $f''(z) > 0$.

<u>Proof</u>: if $f''(\xi) \leq 0 \quad \forall\, \xi \in (y_1; y_2)$, then f' would be decreasing and

$$f'(y_2) \leq f'(y_1)$$

in contradiction with the results of steps 1, 2. $\boxtimes$

7) a) i) We integrate by parts

$$\int_1^{e^2} x^4 \cdot \log x \, dx = \dots$$

$$\begin{bmatrix} u' = x^4 \longrightarrow u = x^5/5 \\ v = \log x \longrightarrow v' = 1/x \end{bmatrix}$$

$$\dots = \frac{x^5}{5} \cdot \log x \Big|_1^{e^2} - \int_1^{e^2} \frac{x^5}{5} \cdot \frac{1}{x} \, dx$$

$$= 2\frac{e^{10}}{5} - \frac{1}{25} x^5 \Big|_1^{e^2} = \frac{1}{25}(9e^2 + 1)$$

ii) $\displaystyle\int_0^{\log 10} \sqrt{e^x - 1} \, dx = \dots$ $\quad z = \sqrt{e^x - 1}$

$$\Rightarrow e^x = z^2 + 1$$

$$\Rightarrow x = \log(z^2+1)$$

$$\Rightarrow dx = \frac{2z}{z^2+1} dz$$

$$\dots = \int_0^3 \frac{2z^2}{z^2+1} dz$$

$$= 2 \cdot \int_0^3 \left(1 - \frac{1}{z^2+1}\right) dz = 2 \cdot (z - \operatorname{arctg} z) \Big|_0^3$$

$$= 2 \cdot (3 - \operatorname{arctg} 3)$$

iii) $\displaystyle\int_0^{\pi/2} \frac{\sin x \cdot \cos x}{1 + \sin^2 x} dx = \dots$ $\quad z = 1 + \sin^2 x$

$$dz = 2 \sin x \cdot \cos x \, dx$$

$$\dots = \frac{1}{2} \int_1^2 \frac{1}{z} dz = \frac{1}{2} \log z \Big|_1^2 = \frac{1}{2} \cdot \log 2$$

b) i) $\displaystyle\int_0^1 \frac{1}{x^{15}} \cdot e^{-1/x}\, dx = \lim_{\varepsilon \downarrow 0} \int_\varepsilon^1 \frac{1}{x^{15}} e^{-1/x}\, dx$

$= \cdots \quad y = 1/x$

$dy = -1/x^2\, dx$

[Here, it's enough to notice that $\displaystyle\lim_{x \to 0^+} \frac{1}{x^{15}} e^{-1/x} = 0$]

$\displaystyle \cdots = \lim_{\varepsilon \downarrow 0} \int_1^{1/\varepsilon} y^{13} \cdot e^{-y}\, dy = \int_1^\infty y^{13} e^{-y}\, dy$

$y^{13} \leq C \cdot e^{y/2}$ for a sufficiently large $C > 0$.

$\Rightarrow y^{13} \cdot e^{-y} \leq C \cdot e^{-y/2}$

Since $\displaystyle\int_1^\infty e^{-y/2}\, dy < +\infty$

the improper integral under consideration converges.

ii) We have

$$\frac{1}{\log x} \cdot \left[\sqrt{1+x^2} - x\right] = \frac{1}{\log x} \cdot \frac{1}{\sqrt{1+x^2} + x}$$

$$\geq \frac{1}{3x \cdot \log x} \quad \text{because} \quad \sqrt{1+x^2} \leq 1 + x \leq 2x$$

Since $\displaystyle\int_2^\infty \frac{1}{x \log x}\, dx = \lim_{k \to \infty} \log(\log x)\Big|_2^k = +\infty$

the integral under consideration diverges.

8) a) <u>Step 1</u>: let x_n be a sequence in $(0;1]$ with $x_n \to 0$ as $n \to \infty$. Then $f(x_n)$ converges.

<u>Proof</u>: We show that $f(x_n)$ is a Cauchy seq.
Fix $\varepsilon > 0$. Since f is uniformly continuous,
$\exists \delta > 0 : |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.
Since $x_n \to 0$, x_n is a Cauchy seq.
$\Rightarrow \exists N \in \mathbb{N} : n, m > N \Rightarrow |x_n - x_m| < \delta$
Hence: $|f(x_n) - f(x_m)| < \varepsilon \quad \forall n, m > N$
Thus, $f(x_n)$ is Cauchy and it converges. $\blacksquare$

<u>Step 2</u>: if x_n, y_n are two sequences in $(0;1]$ with $x_n, y_n \to 0$, then
$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} f(y_n).$$

<u>Proof</u>: define
$$z_n = \begin{cases} x_n, & \text{if } n \text{ even} \\ y_n, & \text{if } n \text{ odd} \end{cases}$$
Then $z_n \to 0$. From Step 1, $f(z_n)$ converges.
Since
$$f(z_n) = \begin{cases} f(x_n), & \text{if } n \text{ even} \\ f(y_n), & \text{if } n \text{ odd} \end{cases}$$
This implies that $\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} f(y_n)$. $\blacksquare$

Step 1 and Step 2 imply that $\exists\, c \in \mathbb{R}$ with

$$\lim_{n\to\infty} f(x_n) = c$$

for every sequence $x_n \in (0;1]$ with $x_n \to 0$.
This means that

$$\lim_{x \downarrow 0} f(x) = c$$

b) Take $f(x) = 1/x$. Then f is continuous on $(0;1]$, but

$$\lim_{x \downarrow 0} f(x) = +\infty.$$